2 **(a)** State the two conditions necessary for a system to be in equilibrium.

1 ……

……..

2 ……

……... [2]

(b) A rigid buoy is connected via an elastic rope of elastic constant 500 N m^{-1} vertically to the sea bed. When rope is vertical and buoy is in equilibrium, the rope has an extension of 1.80 m.

Horizontal currents in the sea water travelling in the direction of left to right then cause the buoy to be displaced so that the elastic rope makes an angle of θ with the seabed. The elastic rope also extends further by 0.70 m.

(i) Sketch a free body diagram of all the forces acting on the buoy when it is subjected to the sea currents on Fig. 2.1. Label the angle θ and forces clearly.

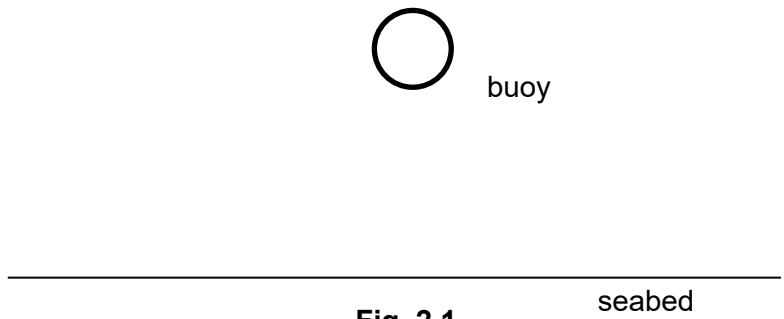

Fig. 2.1

[2]

(ii) Determine the magnitude of the force exerted by the elastic rope on the buoy when there are currents acting on it.

force = …… N [1]

(iii) Determine the resultant of the weight and upthrust acting on the buoy. Show your working clearly

force = …… N [2]

(iv) Hence, determine the magnitude of the force exerted by the currents on the buoy.

force = …… N [1]

(c) Using your values in **(b),** determine the angle θ.

angle = ……° [2]